σ^*

The Resolution Calibration Principle: A Certified Bandwidth for Kernel Fields

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Abstract

A prediction that aggregates training points with distance-decaying weights depends, in principle, on every point. We give a label-free certificate that it depends only on the near ones: past a chosen radius d, the total influence of all sources beyond d is provably below a budget ε. For a *kernel field*—any superposition of distance-decaying kernels of common bandwidth σ—this yields a closed form, $\sigma^* = d/\sqrt{2\ln(A/\varepsilon)}$, with the interfering mass *A measured* from the field, not tuned. The mass is nondecreasing in σ, so one measurement is provably conservative. The same inequality certifies safety, decision stability, abstention, and data locality—the output is ε-insensitive to deleting or corrupting any far point. On frozen ResNet-18, ResNet-50, and ViT-B/16 fields over CIFAR-100, and on text and tabular fields, σ^* holds the budget where naive, plug-in, and density-estimation bandwidths fail by two to five orders of magnitude, coming within *1–5* points of a labelled grid search (about *6* on low-dimensional tabular data) using zero labels, and re-certifies as the field grows. The construction needs only a monotone tail, not the Gaussian: a Laplace field calibrates at parity, and the conservativeness holds for every kernel whose log is concave in log-distance. Softmax attention and prototypical networks reduce to kernel fields, so the certificate transfers to both—fully for prototypes, whose source weights are bounded, and as the interference-tail bound for attention. The bandwidth question, for a kernel field, is a calibration, not a search.

Introduction

A *kernel field* is one of the most reused constructions in applied mathematics: place weighted sources at points z_i of a metric space, attach to each a Gaussian kernel of common width σ, and read the field at any query y as the superposition of the nearby kernels. Kernel density estimators, Nadaraya–Watson regressors, radial-basis interpolants (whose shape parameter is the same bandwidth choice), Gaussian-process posteriors, and the local correction fields that continually adapt a frozen model are all instances of this one object. They share one free parameter—the bandwidth σ, which sets the resolution at which the field reads the space. Too fine and the field responds only at its sources; too coarse and it bleeds across the space and washes out the structure it was meant to resolve.

How is σ chosen? Almost always by search: a grid of candidate values, a random sample, or a cross-validated sweep, each scored by instantiating the field and measuring an outcome. Every such method spends an *outcome signal*—a held-out likelihood, a validation loss, a labelled accuracy—at every candidate, and none of them returns a guarantee about the value it selects.

This paper shows that for a kernel field the search is unnecessary, and that it asks the wrong question. The right question is not ``which σ scores best on held-out labels" but ``which σ keeps the field's cross-source interference below a stated budget." That question has a closed-form, label-free answer that carries a certificate a labelled search cannot: the *certified frontier* $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$ is the largest bandwidth whose aggregate interference past a radius d is provably below ε, with the effective mass *A measured* from the field, not tuned (Section 2). Its structural core is that this measured mass is nondecreasing in σ, so a single measurement is provably conservative; and the one interference bound reads at once as a safety, a performance, an abstention, and a *locality* certificate—the last a label-free guarantee that the prediction depends only on the training points within d, which every distance-based predictor can in principle state (Section 5). On three frozen backbones and four further modalities σ^* holds the budget where every label-free rule fails, matches a labelled grid search with zero labels, and re-certifies as the field grows (Sections 3–4).

The Certificate

Kernel fields.

Fix a metric space, taken here as R^n. A *kernel field* is a finite collection $F=(z_i,v_i)_{i=1}^M$ of sources $z_i\in R^n$ carrying signed weights $v_i\in R$, together with a Gaussian kernel of common bandwidth σ, written in squared distance as

$$K_\sigma(t)=\exp(-t/2\sigma^2), t=\|y-z_i\|^2$$

The field's value at a query y is the superposition $\Sigma_i v_i K_\sigma(\| y-z_i\|^2)$, optionally normalised. Density estimators, interpolants, and the correction fields of Section 3 are all fields of this form. Write $V_{max}=max_i|v_i|$; the certified quantity is invariant under $(v_i,\varepsilon)(cv_i,c\varepsilon)$, so we fix $V_{max}=1$ (the budget ε is in units of the largest source weight), which holds exactly for the one-hot Parzen vote of Section 3.

Near and far, and the effective mass.

Around a query y, a separation radius d partitions the sources: those within d are *near* (the field should respond to them), those in $F=i:\| y-z_i\|>d$ are *far* (it should leave them undisturbed). The far tail $Tail_\sigma(y,d)=\Sigma_{i\in F}|v_i|\, K_\sigma(\| y-z_i\|^2)$ is the field mass leaking past d; the bandwidth question is how large σ can be before it exceeds a budget ε. Not every far source matters equally: most are negligibly weak, and a handful near the radius carry the tail. The right count is therefore not $|F|$ but the number of far sources that *effectively* contribute—the participation ratio of the far weights $\omega_i=K_\sigma(\| y-z_i\|^2)$, which equals the true count when they are equal and drops toward one when a few dominate,

$$N_{\mathrm{eff}}(\sigma)=\frac{\left(\sum_{i\in\mathcal{F}}\omega_i\right)^2}{\sum_{i\in\mathcal{F}}\omega_i^2}\in[1,|\mathcal{F}|]$$

which must be restricted to F: a participation ratio over all sources is dominated by the near mass and does not bound the tail. The field mass acting past the radius is $A=V_{max}N_{eff}\geq 1$.

Assumptions.

The certificate rests on three explicit conditions. (A1) *Bounded weights*: $|v_i|\leq V_{max}=1$. (A2) *A meaningful near/far radius*: $d>0$ partitions sources into a near set the field should respond to and a far set it should not, so every far source satisfies $\omega_i\leq K_\sigma(d^2)$. (A3) *Budget* *below mass*: $0<\varepsilon<A$. It is worth separating three kinds of statement in what follows: the frontier and the safety, stability, and conservativeness results are *proven* under (A1)–(A3); the effective mass A and the tail ratios are *measured* from the field; and the radius d (here the tenth percentile of pairwise centre distances) is a *design choice*—any fixed rule gives a valid certificate.

The safety frontier.

Define the leakage certificate $cert(A,\sigma,d)=A\,K_{\_}(d^2)=A\,e^{-d^2/2\sigma^2}$.

Proposition 1 (Safety frontier). *For $A>0$, $d>0$, $0<\varepsilon<A$, the equation $cert(A,\sigma,d)=\varepsilon$ has the unique solution*

$$\sigma^*(A,\varepsilon,d)=\frac{d}{\sqrt{2\ln(A/\varepsilon)}}$$

and the safe set $\sigma>0{:}cert\le\varepsilon$ is $(0,\sigma^*]$.

Sketch. $cert$ is continuous and strictly increasing in σ, rising from 0 to A; solving $cert=\varepsilon$ gives the unique frontier, and monotonicity makes the safe region $(0,\sigma^*]$ (SI Note 1). The single-source bound extends to the whole tail: because every far source lies beyond d, each far kernel is bounded by $K_\sigma(d^2)$, and the participation-ratio mass converts this per-source bound into an aggregate one,

$$\mathrm{Tail}_\sigma(y,d)=\sum_{i\in\mathcal{F}}|v_i|\omega_i\le V_{\max}\,N_{\mathrm{eff}}K_\sigma(d^2)=\mathrm{cert}(A,\sigma,d)$$

with $A=V_{max}N_{eff}$; at $\sigma=\sigma^*$ the aggregate leakage is at most ε (SI Note 1).

The measured mass is self-consistent.

The frontier depends on $A=V_{max}N_{eff}(\sigma)$, but N_{eff} itself depends on σ, so A must be measured at some σ and then used to set σ^*. We measure it once at the pairwise scale $\sigma_{pair}=d/\sqrt{(2\ln(1/\varepsilon))}$ (the frontier for a single far source) and deploy at σ^*. This one-shot procedure is provably conservative:

Proposition 2 (Conservativeness). *$N_{eff}(\sigma)$ is nondecreasing in σ. Consequently $\sigma^*\le\sigma_{pair}$, the mass measured at σ_{pair} satisfies $A(\sigma_{pair})\ge A(\sigma^*)$, and the true tail at the deployed bandwidth obeys $Tail_{\sigma^*}(y,d)\le A(\sigma_{pair})\,K_\sigma[\sigma^*](d^2)=\varepsilon$.*

Sketch. As σ grows the far weights become more uniform, so their participation ratio cannot decrease; since $\sigma_{pair}\ge\sigma^*$, measuring the mass there overestimates the deployed mass, and the one-shot certificate holds (SI Note 2). On the three frozen backbones the one-shot mass over-estimates the deployed mass by *2.2–4.3×* (Fig. 1a) and the measured one-shot tail is *0.08–0.16* ε—strictly inside budget on every network. Iterating $A\to\sigma^*\to A$ converges to a fixed point in six to seven steps (Fig. 1b), but the one-shot value already certifies. Both load-bearing facts—the participation-ratio tail bound and this monotonicity—are confirmed independently on random configurations (zero violations in *20,000* and *3,000* trials respectively; SI Note 8).

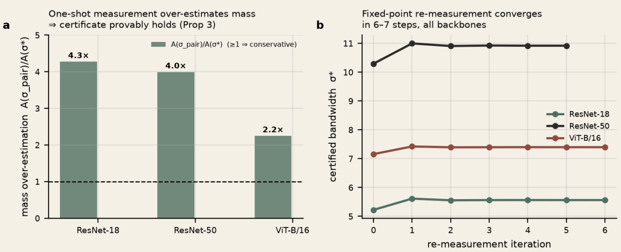

Figure 1. The measured mass is conservative. (a) Measuring A once at σ_{pair} over-estimates the mass at σ^* by *2.2–4.3×* on all three backbones, so the one-shot certificate provably holds (Proposition 2). (b) Iterating the re-measurement converges to a fixed point in *6–7* steps.

The same bound controls the decision.

Consider a field used as a classifier: each class c receives a score $S_c(y)=\Sigma_{i:\,class(i)=c}\,v_i\,K_\sigma(||\,y-z_i||^2)$, with prediction $_c\,S_c$. Splitting each score by the near/far partition, the certificate bounds the far part of every class score by ε at $\sigma=\sigma^*$. Let the *local prediction* be $_c\,S_c^{near}$ and the *local margin m(y)* the gap between the top two near-scores.

Theorem 1 (Decision stability). *If $\sigma\le\sigma^*$ and $m(y)>2\varepsilon$, the deployed prediction equals the local prediction: the far field cannot change the decision.*

Sketch. Each class score moves by at most ε under far-field interference, so a two-class gap moves by at most 2ε; a near margin above 2ε cannot be flipped (SI Note 3). This is the perturbation bound underlying interval-based certified robustness ; what is new is *where* ε *comes from*—a bandwidth calibrated so the far-field perturbation is provably at most ε, read from geometry without labels. The test uses only near scores and ε, so a deployer flags certified queries per query, and a floor follows:

Corollary 1 (Performance floor). *At $\sigma\le\sigma^*$, deployed accuracy $\ge$ (near-field accuracy) - [$m(y)\le 2\varepsilon$].*

The same inequality certifies *data locality*. Let $F'(y)$ be the field with its far set F arbitrarily deleted, or replaced by any sources of bounded weight $|v_i'|\le V_{max}$ located outside the radius d.

Corollary 2 (Far-set locality). *At $\sigma\le\sigma^*$: (i) deleting the far set moves the output by at most the budget, $||\,F(y)-S(y)||=||\,T(y)||\le\varepsilon$; (ii) replacing it by any admissible far set shifts each class score by at most ε, so $||\,F'(y)-F(y)||\le2\varepsilon$; and (iii) if $m(y)>4\varepsilon$ the decision is unchanged under any such deletion or corruption. The prediction is therefore ε-insensitive to the presence, and 2ε-insensitive to the identity, of every training point beyond d.*

Sketch. (i) is Proposition 1 verbatim: $Tail_\sigma(y,d)\le\varepsilon$ *is* $||\,T(y)||\le\varepsilon$. For (ii) each admissible far set contributes at most $V_{max}N_{eff}K_\sigma(d^2)=\varepsilon$ to any class score by the same certificate, so the original and corrupted far contributions each lie in a ball of radius ε and differ by at most 2ε; (iii) is then Theorem 1 applied to a 2ε perturbation. All three hold empirically on four datasets, and fail by two to three orders of magnitude at a bandwidth *3×* too wide (Fig. 2).

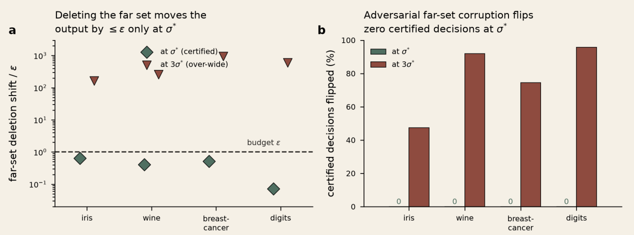

Figure 2. Certified data locality (Corollary 2). (a) At σ^* deleting the entire far set moves the output by ≤ε on all four datasets; a bandwidth *3×* too wide overshoots by two to three orders of magnitude. (b) Under worst-case adversarial corruption of the far set, zero certified-stable decisions flip at σ^*, versus *47–96%* at $3\sigma^*$.

One certificate, four readouts.

At $\sigma=\sigma^*$ every query splits the field as $F(y)=S(y)+T(y)$ with $||\,T(y)||\le\varepsilon$—which *is* Proposition 1. Because $S(y)$ is precisely the response the field would give with the entire far set deleted, the same inequality reads four ways:

- *Safety*: the deployed response is within ε of the near-field response, $||\,F(y)-S(y)||\le\varepsilon$.
- *Performance*: if the near margin exceeds 2ε, $F=S$ (Theorem 1).
- *Abstention*: if the top near score is $\le2\varepsilon$ the certificate cannot fire, and the query lies outside the field's certified support—a label-free per-query support signal.
- *Locality*: the output is ε-insensitive to the presence, removal, or corruption of *every* far source—deleting the entire far set moves

$F(y)$ by at most ε, and no admissible far set can flip a decision with near margin above $4ε$ (Corollary 2). This is a measured, per-configuration data-locality guarantee: it certifies *which* training points a prediction can depend on, and connects the bandwidth to unlearning and attribution rather than to density estimation. It is not a differential-privacy bound—there is no randomisation and the guarantee is over the field as measured, not worst-case over datasets—but it occupies the same role for distance-based predictors.

We confirmed the decomposition on *20,000* certified-regime configurations with zero violations, and on a real frozen ResNet-18 field (*20,000* kernels, *10,000* queries) the certified σ^* flags *27%* of queries stable with a stable-implies-equal rate of *100.00%*—zero violations—while the naive bandwidth certifies nothing and its accuracy has collapsed. The certified-stable set sits on the field's accuracy–coverage curve rather than above it; its distinguishing property is that the selection rule is label-free and a priori (SI Note 6). A Lipschitz argument extends the pointwise bound to a ball: the certified radius is genuine but small ($\delta \approx 1.6$–2.2% of inter-query distance), so the per-query certificate carries the deployment claims (SI Note 4).

The Flagship: Three Frozen Backbones

The certificate was built for the dense regime, where many sources act at once and the naive single-interferer intuition fails. We reproduce that failure, and its cure, on real frozen deep networks.

Construction.

For each backbone—ResNet-18 (*512*-d), ResNet-50 (*2048*-d), and ViT-B/16 (*768*-d), all ImageNet-pretrained and frozen—we extract the penultimate features of CIFAR-100 and build a correction field of *20,000* balanced kernels (*200* per class) over the training features, classified by a kernel-weighted (Parzen) vote. The locality radius d is the tenth percentile of pairwise centre distances—pure geometry, no labels. We fix $ε=0.05$ a priori and compare three bandwidths: the naive single-interferer rule $\sigma_{pair}=d/\sqrt{2\ln(1/ε)}$; the certified frontier σ^*, whose mass $A=V_{max}N_{eff}$ is *measured* once at σ_{pair}; and, as an oracle reference, a labelled grid search over a *30*-point log-spaced grid (denoted ``grid"). We also run the classical label-free density bandwidths—Silverman, Scott, and Abramson's square-root law.

Computing σ^* (label-free).

Four steps, no labels: (1) fix ε and set d to the tenth percentile of pairwise centre distances; (2) form $\sigma_{pair}=d/\sqrt{2\ln(1/ε)}$; (3) measure $A=V_{max}N_{eff}$ at σ_{pair}, averaging the far-mask participation ratio over queries; (4) return $\sigma^*=d/\sqrt{2\ln(A/ε)}$. By Proposition 2 measuring at $\sigma_{pair}\geq\sigma^*$ over-estimates A, so one pass suffices. Full computational specification, grids, and seeds are in SI Note 7 and the repository.

Results.

The pattern is identical across all three architectures (Fig. 3, Table 1). The naive bandwidth overshoots the interference budget by three to four orders of magnitude (tail/ε from 4.9×10^3 to 1.0×10^4) and collapses the field into a single component spanning 97–100% of the centres. The certified σ^* holds the budget (tail/ε between *0.08* and *0.16*), dissolves the giant component to 4–7%, and recovers accuracy to within 1–5 points of the labelled grid-search optimum—using *zero labels*, with ε fixed across every backbone so the adaptation to *512*, *768*, or *2048* dimensions happens automatically through the measured mass A. The classical selectors are not straw comparators and fail here: Silverman and Scott over-smooth catastrophically (tail/ε$>10^5$, accuracy 27–64%, *below* the naive rule), Abramson's rule worse still (12–36%). The two label-free distance heuristics—the median heuristic and a k-nearest-neighbour scale, the natural competitors to σ^*—pick a bandwidth *3.6–6*× larger, which drives tail/ε to 10^5 and loses *9–26* accuracy points: the quantile d fixes the *scale*, but the logarithmic factor $\sqrt{2\ln(A/ε)}$ is what turns it into a *safe* one, and only the calibration supplies it. Over five re-samples the certified accuracy varies by *≤0.3* points on every backbone (*56.3±0.2*, *51.1±0.2*, *72.2±0.2 %*), so the ordering is not seed-luck.

The certificate is robust to its own design choices.

Four checks, all label-free, confirm the operating point is not fragile (SI Note 5). *From average to per-query.* The mass A is a query-average, so the certificate is stated on the average tail; in deployment one cares about the *worst single query*. Empirically the two coincide: over *10,000* ResNet-18 queries the largest single-query tail/ε is *0.24*, still well inside budget, so no query exceeds it. To make the per-query guarantee a priori rather than empirical, set A to an upper quantile of N_{eff} instead of its mean; this costs *≤0.2* points of accuracy and gives zero exceedances on all three backbones. The near set is *25–47*× class-enriched (weighted near purity *0.31/0.25/0.47* against a chance level of *0.01*), though this is a *plurality*: the certificate guarantees *which* sources decide, not that they decide correctly, and accuracy remains a separately measured outcome. And accuracy stays flat across the ε×percentile grid (*≤6.7* points, over a realized σ^* range of only *1.34–1.41*×; SI Note 5) while the certified tail stays in budget—a conjunction that a merely accuracy-tuned dial would not show, since a bandwidth *4–6*× larger fails catastrophically.

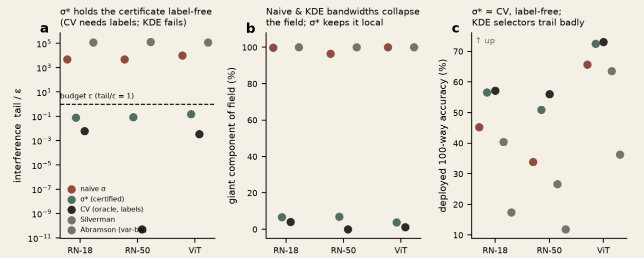

Figure 3. Certified σ^* across three frozen backbones (CIFAR-100, $ε=0.05$, zero labels). (a) Only σ^* holds the interference budget (tail/$ε<1$); naive and classical KDE bandwidths overshoot by 10^3–10^5×. (b) Naive and KDE bandwidths collapse the field into one component; σ^* keeps it local. (c) σ^* matches the labelled grid-search accuracy label-free, while the KDE selectors trail badly.

Backbone	Method	tail/ε	giant%	acc%
ResNet-18	naive	5068	100	45.2
(512d)	median heur.	215695	100	39.6
	k-NN(7)	120953	100	40.5
	σ*	0.08	7	56.6
	grid (labels)	0.01	4	57.2
	Silverman	125520	100	40.4
ResNet-50	naive	4892	97	33.9
(2048d)	median heur.	210717	100	25.3
	k-NN(7)	129605	100	26.6
	σ*	0.09	7	51.0
	grid (labels)	0.00	0	56.1
	Silverman	127973	100	26.6
ViT-B/16	naive	9963	100	65.7
(768d)	median heur.	216464	100	63.1
	k-NN(7)	110309	100	63.7
	σ*	0.16	4	72.5
	grid (labels)	0.00	1	73.0
	Silverman	123721	100	63.6

Table 1. Certified σ^* versus naive, median-heuristic, k-nearest-neighbour, labelled grid-search, and classical density bandwidths on three frozen backbones. The two label-free distance heuristics—the natural competitors to

σ^*—pick a bandwidth *3.6–6*× too large, overshooting the interference budget by roughly five orders of magnitude and losing *9–26* accuracy points; σ^* is the only label-free rule that holds tail ≤ε and stays near the labelled grid-search accuracy.

Beyond vision: four non-image modalities.

The certificate is a property of the kernel-field geometry, not of the image domain. We test it on four corpora unrelated to CIFAR (Table 2, Fig. 4), each over five seeds: three text collections—DBpedia, AG-News, Yahoo-Answers—embedded with a frozen sentence transformer (*384*-d), and a raw tabular set (Covertype, *54*-d). The naive bandwidth overshoots by 3.6×10^2 to 2.3×10^3 while σ^* holds tail/ε<*1*, and every stable-flagged query has its deployed prediction equal to the intended one (*100%* on every corpus). On the three high-dimensional text sets σ^* matches the labelled grid search to within *0.9* points *label-free*; on the low-dimensional tabular set it trails by *5.8* points, the same specificity-first behaviour as the public datasets. The pattern the flagship establishes on vision reproduces, with error bars, on text and tabular data alike.

Corpus (dim)	method	tail/ε	acc%
DBpedia-14	σ*	0.24	92.1,±0.6
text (384)	grid (lab.)	---	92.3,±0.5
	median	45609	89.0,±0.7
	k-NN(7)	33492	89.1,±0.6
AG-News-4	σ*	0.27	87.9,±1.1
text (384)	grid (lab.)	---	88.0,±1.1
	median	13018	85.7,±0.8
	k-NN(7)	10272	85.8,±0.9
Yahoo-10	σ*	0.30	66.9,±0.5
text (384)	grid (lab.)	---	67.8,±0.3
	median	32631	65.8,±0.4
	k-NN(7)	24822	66.0,±0.4
Covertype-7	σ*	0.10	65.0,±0.8
tab. (54)	grid (lab.)	---	70.8,±1.1
	median	20755	53.9,±1.5
	k-NN(7)	102	60.0,±1.6

Table 2. The certificate generalises across modality (5 seeds, ε=*0.05*; naive omitted for space, tail/ε 3.6×10^2–2.3×10^3). σ^* holds tail/ε<*1* label-free and matches the labelled grid search within *0.9* points in high dimension; the *5.8*-point trail on low-dimensional tabular data is the same specificity-first regime as Section 4. The label-free median and *k*-NN heuristics overshoot the budget by 10^2–10^5× on every corpus (the same failure as Table 1), with an accuracy cost of *1–3* points on text and up to *11* on tabular. The stable-implies-equal rate is *100%* throughout.

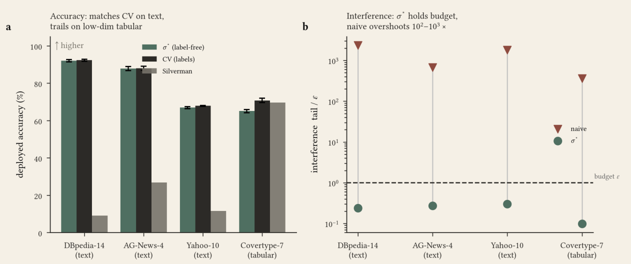

Figure 4. Across four non-image modalities. (a) σ^* (label-free) matches the labelled grid search on the three text sets and trails only on low-dimensional tabular data; error bars are ±*1* s.d. over five seeds. (b) σ^* holds the interference budget on every corpus, while the naive bandwidth overshoots by two to three orders of magnitude.

Beyond the Gaussian: any monotone tail.

Nothing in the certificate uses the Gaussian shape—only that the far weight $\omega_i=K_\sigma(\|y-z_i\|)$ decays monotonically in distance and that the far participation ratio flattens as σ grows. For any tail K_σ the frontier is the σ solving $A\,K_\sigma(d)=\varepsilon$; for the Laplace kernel $K_\sigma(r)=e^{-r/\sigma}$ this gives $\sigma^*=d/\ln(A/\varepsilon)$—the same radius-over-a-log-factor structure, with the square-root gone because the tail is exponential in *r* rather than r^2. The conservativeness argument (Proposition 2) transfers verbatim: the far weights still become more uniform as σ grows, so N_{eff} is nondecreasing and one measurement still over-estimates. We verify this on iris, wine, breast-cancer, and digits (Fig. 5): the Laplace σ^* holds tail/ε<*1* on all four (mean *0.16–0.24*), matches a labelled grid search anchored to its own scale within *0–2.8* points, and shows zero monotonicity violations—exactly the Gaussian behaviour under a different tail. The principle is a property of monotone-tail kernel fields, not of the Gaussian.

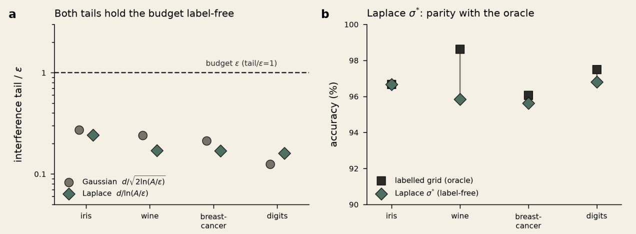

Figure 5. The calibration is not Gaussian-specific. On four public datasets the Laplace frontier $\sigma^*=d/\ln(A/\varepsilon)$ behaves like the Gaussian one. (a) Both tails hold the interference budget label-free (tail/ε<*1*). (b) The Laplace σ^* reaches parity with a labelled grid search anchored to the Laplace scale.

Beyond kernel density: a transformer's attention.

The predictors above are all built as kernel fields by construction. Softmax attention is not—its weight on key *i* is $(q,k_i/)$, an inner product, not a distance. Yet the polarisation identity $q,k_i=12(\|q\|^2+\|k_i\|^2-\|q-k_i\|^2)$ makes the query-only factor cancel in the softmax, leaving *exactly* a Gaussian kernel field over the keys with bandwidth $\sigma=\sqrt{()}$ and per-source weight $v_i=e^{\|k■\|^2/2}$ (SI Note 9). The certificate therefore applies verbatim, with certified temperature $^*=(\sigma^*)^2=d^2/(2\ln(A/\varepsilon))$. We test this on the real query/key vectors of every sampled head of two pretrained transformers—BERT-base and GPT-2—over long contexts (*360* keys per head). The identity is exact to the models' float32 precision ($<10^{-4}$ across all heads), and at * the certified far-key interference tail stays within budget across *35,000* queries per model with *zero* violations, overshooting by *18–33*× when is pushed to 4^* (Fig. 6). Attention's source weights v_i are unbounded, so this is the interference-tail certificate (Proposition 2) rather than the full $V_{max}=1$ locality readout, which transfers cleanly only when key norms are bounded—the pure-Gaussian case (the measured normalised far mass at * is *0.35* for BERT and *0.62* for GPT-2, not ε-small, but well below the *0.80–0.84* at 4^*). The reduction is a theorem, not a conjecture: a dot-product readout the certificate was not built for turns out to be one of its instances.

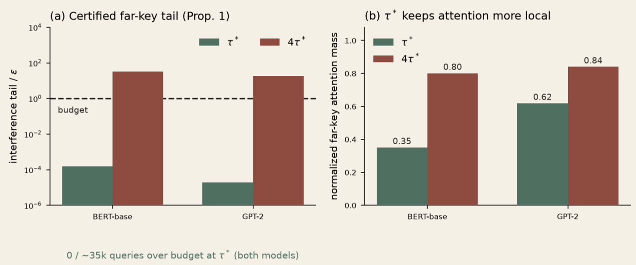

Figure 6. Attention inherits the certificate. On real query/key vectors from BERT-base and GPT-2 (all sampled heads, *360* keys each). (a) At the certified temperature * the far-key interference tail is far below budget (mean $<10^{-3}\varepsilon$; zero of *35,000* queries per model exceed it) and blows up *18–33*× at 4^*. (b) * holds attention markedly more local than an over-wide temperature.

The bounded-weight companion: prototypical networks.

Attention exercises the unbounded-weight edge of the class; a prototypical network exercises the clean interior. It classifies a query by $softmax_k(-\|f(x)-c_k\|^2/)$ over class prototypes c_k—already a Gaussian kernel field, with *no* polarisation needed and all source weights $v_k=1$, so assumption (A1) holds with $V_{max}=1$ exactly. Both the interference-tail certificate *and* the full far-set locality readout (Corollary 2) therefore transfer. We train a *4*-layer convolutional ProtoNet on Omniglot (test accuracy 91.5%) and certify its prototype fields on held-out *60*-way episodes (*12,000* queries). The identity is exact to float32 ($<10^{-6}$); at the certified σ^* the far-prototype tail sits at 0.25ε with *zero* violations, and—because the weights are bounded—the normalised far mass is a genuinely ε-small *0.08*, against attention's *0.35–0.62* (Fig. 7). Attention and prototypes thus bracket the two ends of the same class: the tail bound holds throughout, and the full locality guarantee holds wherever weights are bounded.

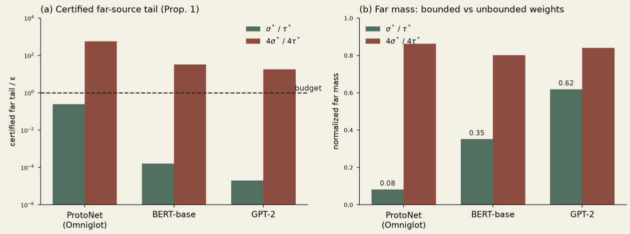

Figure 7. A family of readouts inherits the certificate. Prototypical networks (Omniglot) alongside the two transformers. (a) Every architecture holds the certified far-source tail below budget at its own σ^*/τ^* and overshoots at *4*× (*18–33*× for attention, *560*× for ProtoNet). (b) The normalised far mass separates the bounded case (ProtoNet, $v_k=1$, far mass *0.08*—ε-small) from the unbounded case (attention, *0.35–0.62*): the full locality readout transfers exactly only when weights are bounded.

The Budget Travels

A labelled search fixes σ once, on the field it was tuned on. A deployed correction field grows—edits accumulate, classes are added—and the interference structure changes underneath the fixed choice. Because σ^* re-measures *A*, it re-certifies for free; a labelled grid search cannot.

We grow the ResNet-18 field by density, from *3* to *120* examples per class (*300* to *12,000* centres), and fix the labelled grid-search $\sigma=8.83$ once on the sparse initial field. As the field densifies that fixed width becomes an over-smoother: its interference climbs from tail/$\varepsilon=12$ to *502* (Fig. 8b) and its accuracy reaches only 47.1% at full density, while σ^*—re-measuring *A* and shrinking from *6.51* to *5.41*, holding tail/ε near *0.10*—reaches 54.1%, a *7*-point margin with *zero* labels (Fig. 8a). This margin is against a bandwidth *deliberately frozen* on the sparse field, which is the failure mode we demonstrate—not against a labelled search re-run at each density. Against that per-stage re-tuned oracle σ^* does not dominate; it *tracks* it to within a fraction of a point at every density, label-free. The claim is about adaptivity and the cost of not re-tuning: the labelled search is optimal for the field it saw and silently wrong for the field it is deployed on.

Public datasets.

On iris, wine, breast-cancer, and digits the naive bandwidth overshoots by 2.8–147× while σ^* holds tail/$\varepsilon<1$. These fields are neither dense nor growing, so the labelled grid search is a fair fixed comparator, and σ^* matches it within about two points on three of four (breast-cancer 96.6 vs 96.5, digits 97.0 vs 97.0, iris 96.3 vs 96.3%); on wine it trails by 2.2 points. σ^* is a specificity-first bandwidth: at fixed low density it delivers parity, and in the dense, growing regime the guarantee turns into an accuracy win.

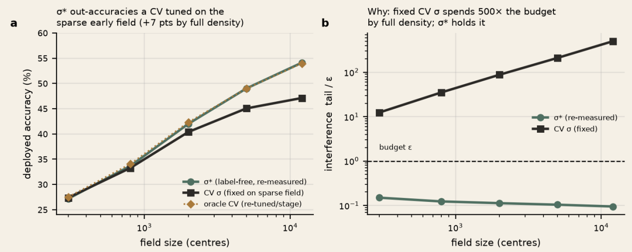

Figure 8. The budget travels—and buys accuracy. A labelled grid-search σ is fixed once on the sparse *3*-per-class field. (a) As the field densifies, σ^* (label-free, re-measured) exceeds the fixed grid-search σ by *7* points at full density and tracks a per-stage re-tuned oracle. (b) The fixed bandwidth's interference climbs to *500*× the budget while σ^* holds tail ≈*0.10*ε.

Related Work

Density-estimation rules—plug-in bandwidths (Silverman, Scott), Sheather–Jones , variable-bandwidth and diffusion selectors , Lepski's method , and scale-space —target an integrated squared error against a true *density*; we target *interference*, and derive the bandwidth from that objective using no labelled signal. Run on the three frozen backbones, these objective-mismatched selectors miss the budget in *opposite* directions: Sheather–Jones under-smooths to a near-delta kernel (accuracy at chance), while Gaussian-process marginal-likelihood over-smooths to *11–15*× σ^* (tail/$\varepsilon 3\times10^5$)—confirming that likelihood- and ISE-driven selection is blind to the interference budget in either direction. The label-free distance heuristics—the median heuristic and local scaling —fix the scale but not the safety factor, and overshoot the budget by roughly five orders of magnitude on the dense fields (Table 1). Gaussian-process length scales (learned by the marginal-likelihood optimisation measured above) have motivated uniform error bounds for safety-critical use ; σ^* fixes the scale from geometry and offers an a priori certificate rather than a fitted posterior. Conformal prediction and selective classification certify *marginal* coverage from a *labelled* calibration set, whereas Corollary 1 certifies *per-query decision equality* without labels—both require a representative sample (label-free is not sample-free: *A* is measured over held-out queries), and the two are complementary, a conformal wrapper calibrating the residual far-field probability our margin leaves uncontrolled. In continual learning , where a correction field is adapted over a frozen model, the interference budget makes the stability–plasticity trade-off quantitative. The locality readout (Section 2) relates to machine unlearning and influence-based attribution : those quantify how removing a training point changes the output, while σ^* *certifies*, label-free and a priori, that removing any far point changes the output by at most ε and corrupting it cannot flip a *4*ε-margin decision (Corollary 2)—a measured data-independence guarantee for distance-based predictors, distinct from the randomised, worst-case sensitivity of differential privacy.

Which models inherit the certificate.

The certificate uses only that the output is a weighted sum of training points with a distance-decaying weight, so it is a property of a *family* of modern readouts, not of kernel density estimation alone. We demonstrate two members on real networks, chosen to bracket the class. Softmax attention reduces exactly to a Gaussian kernel field over keys with unbounded source weights; its certified far-key tail holds on BERT and GPT-2 (Fig. 6). Prototypical networks are the bounded-weight companion ($v_k=1$), for which the full locality readout transfers, not just the tail; the certificate holds on a trained ProtoNet with ε-small far mass (Fig. 7). Between them they instantiate both ends of the assumption (A1). Two further families remain candidates we do *not* test here: representer-point decompositions write a network's logit as such a field, and a network near convergence acts as a kernel machine in its neural-tangent space . For each, σ^* would supply a candidate locality radius from the field's own geometry; verifying the budget on them is future work.

Limitations

Three boundaries. On pure density recovery there is no near/far structure—every sample is near—so the separation objective is silent; the principle applies to fields that must *discriminate*. σ^* is specificity-first: its guarantee is on interference and accuracy dominance is regime-dependent (parity at fixed low density, an accuracy win only in the dense, growing regime). And conservativeness (Proposition 2) is stated at fixed radius *d*; *d* drifts by only *1%* over a *40*× densification, consistent with $O_p(n^{-1/2})$ quantile convergence (SI Note 5), so re-estimating it per stage corrects a vanishing term, but a field whose class-conditional geometry itself changes would need *d* re-derived.

Conclusion

A kernel field has one honest bandwidth: the largest σ whose cross-source interference is provably below a stated budget. It has a closed form, $\sigma^*=d/\sqrt{2\ln(A/\varepsilon)}$; it needs no labelled search because the mass *A* is measured from the field; and it carries a certificate that a search cannot—not only that the field is safe, but that every prediction it flags stable is decided by its intended sources. Measuring the mass once is provably conservative, so no tuning is needed, and as the field densifies the same re-measurement exceeds a labelled bandwidth frozen on the sparse early field. The bandwidth question, for a kernel field, is not a search but a calibration: derive the certified bandwidth from geometry, and read off the guarantee.

Data and code availability.

All experiment scripts and released result files are available at `github.com/negulescu42/sigma-star` with fixed seeds and the exact bandwidth grid for each experiment. Every empirical certificate reported here—including the maximum-tail and zero-exceedance claims—is reproducible end-to-end from these scripts on the stated frozen features.

Author contributions.

R.N. conceived the resolution-calibration framing and developed the empirical and machine-learning program (Sections 3–4). C.B. is responsible for the analytic core: the safety frontier (Proposition 1), the conservativeness theorem (Proposition 2), the decision-stability theorem (Theorem 1), the far-set locality corollary (Corollary 2), and the uniform-over-region certificate (SI Note 4). Both authors reviewed and approved the final text.

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